FIRMWARE DOCUMENTATION

superOS-303

Owner's Manual

COMBINED EDITION V1.0-BETA · CUSTOM FIRMWARE FOR THE ROLAND TB-303
VOX DEI CPU · MICHIGAN SYNTH WORKS

CONTENTS

REFERENCE

PRECAUTIONS

Firmware installation and any modification of your instrument are performed entirely at your own risk. Back up your patterns over SysEx before updating. Do not disconnect power during an update unless the update has failed. We are not responsible for damage to your instrument, loss of data, or any reduction in the value of vintage hardware resulting from installation, modification, or use of this firmware.

1 ABOUT SUPEROS-303

superOS-303 is custom firmware for the open-source VOX DEI CPU by **Michigan Synth Works**, a drop-in replacement for the Roland TB-303's original processor. It is a fork of DJ Phazer's **OS-303**.

The goal is to extend the TB-303 sequencer beyond what was possible in 1981: a modern, flexible environment for writing, transforming, and performing patterns, while keeping the instrument's original workflow intact. Source code and releases are on **GitHub**.

The **Combined Edition** holds two complete operating systems in one chip: superOS-303 and **D650C mode**, a cycle-accurate emulation of the original TB-303 CPU running the genuine Roland mask ROM. You can switch between them from the panel at any time; both keep their own pattern memory. See the Combined Edition chapter.

This site also provides a **web pattern editor** that talks to the hardware over MIDI, and a **case customizer** for previewing custom case designs.

2 HOW TO USE THIS MANUAL

A. PANEL NAMES

Controls are written as they appear on the panel: FUNCTION PITCH MODE TIME MODE CLEAR BACK WRITE/NEXT RUN/STOP ACCENT SLIDE DOWN UP, the keyboard keys C through C′ (high C), and the pattern keys 1–8. The **Mode Dial** selects Pattern Write, Pattern Play, Track Write, or Track Play; the **Group Dial** selects pattern group I–IV (and the track number in Track mode).

A combination such as FUNCTION + CLEAR means: hold the first control, then press the second.

> **NOTE**
>
> In Time Mode the four writing keys carry the original panel legends. This manual uses those names: STEP is the DOWN key, TIE is the UP key, REST is the ACCENT key, and LOCK is the SLIDE key.

Everything in this manual describes **superOS mode** unless stated otherwise. In D650C mode the instrument behaves like a stock TB-303; see Combined Edition, section 2.

1 PREPARING A PATTERN

MODE DIAL PATTERN WRITE

A. SELECT A PATTERN

Set the Mode Dial to Pattern Write. Choose a group with the Group Dial and press a pattern key 1–8 to select a slot.

B. CLEAR THE PATTERN

Hold the pattern key and press CLEAR. The LEDs flash briefly to confirm the slot has been erased.

2 SETTING THE LENGTH

MODE DIAL PATTERN WRITE

A pattern section can be 1 to 32 steps long (1 to 24 in triplet mode), and the length can be changed at any time, even while the sequencer is running. Linking a slot's A and B sections plays them as one pattern of up to 64 steps (Playing Patterns, section 5). Length changes are never destructive: notes past the last step are retained, so shortening and then lengthening restores them.

A. COUNT THE STEPS

Hold FUNCTION and tap STEP once per step. The first tap resets the length to 1; every further tap adds one step. Eight taps make an 8-step pattern.

B. SET THE LENGTH DIRECTLY

Hold FUNCTION, press a black key to choose the range, then a white key to choose the exact step within it. The editor opens on the page that contains the current last step, and nothing changes until a white key picks one, so paging around to look is free. While FUNCTION is held, the LEDs display the current length.

BLACK KEY	RANGE	B SECTION (SLIDE ON)
C#	Steps 1–8	Steps 33–40
D#	Steps 9–16	Steps 41–48
F#	Steps 17–24	Steps 49–56
G#	Steps 25–32	Steps 57–64

With an A section selected, FUNCTION + SLIDE moves the editor into the B section (steps 33–64). Picking a step there links the A/B pair automatically, making one long pattern. A B section selected on its own stays capped at 32 steps.

C. ONE STEP AT A TIME

FUNCTION + WRITE/NEXT adds one step; FUNCTION + BACK removes one (down to 1). This works stopped or running.

D. DOUBLE AND HALVE

FUNCTION + UP doubles the length: it either reveals the retained tail or copies the first half into the new second half. Doubling past step 32 spills into the B section and links the pair. FUNCTION + DOWN halves the length (31 → 15 → 7 → 3 → 1) and destroys nothing.

E. TRIPLET MODE

In TIME MODE, FUNCTION + UP switches the section between straight sixteenths and triplet timing (12-step pages, maximum length 24; steps past 24 are retained, not deleted). While FUNCTION is held there, the UP LED shows the current setting.

3 WRITING PITCHES

MODE DIAL PATTERN WRITE · SEQUENCER STOPPED

Press PITCH MODE. Play the keyboard keys in order; each press writes the next note of the pattern.

- Octave: hold DOWN for one octave down, UP for one up, or both together for two octaves up.
- Hold ACCENT and/or SLIDE while pressing a key to write those flags with the note.
- WRITE/NEXT auditions the current step; BACK steps backward.

When every note step has a pitch, the mode exits automatically and the panel returns to normal. To edit pitches later, press PITCH MODE again, or use Step Editing (Pattern Editing, section 1).

4 WRITING RHYTHM

MODE DIAL PATTERN WRITE · SEQUENCER STOPPED

Press TIME MODE. For each step of the pattern, press one key; the cursor advances after each entry. After the last step, the mode exits automatically.

KEY	WRITES
STEP	Note. Plays the next pitch in the pattern.
TIE	Tie. Extends the previous step.
REST	Rest. Silence for one step.
LOCK	Toggles step lock on the current step without advancing. A locked step ignores live edits (Performance, section 1).

NOTE

Pitch and time are independent streams. Changing a step from note to rest does not discard its pitch; the pitch stays in the queue, and writing a note there again plays the same pitch.

5 PLAYING PATTERNS

MODE DIAL **PATTERN PLAY**

Set the Mode Dial to Pattern Play and press **RUN/STOP**. Press a pattern key **1**–**8** to jump to that slot while stopped, or to queue it for the next pattern wrap while running.

A. SECTIONS A AND B

Each pattern slot has two sections of up to 32 steps each. Press **ACCENT** for section A or **SLIDE** for section B. Stopped, the switch is immediate; running, it takes effect at the next wrap. Selecting a section clears any active chain, and selecting one section alone unlinks a linked pair.

B. LINKING A + B: 64-STEP PATTERNS

Hold one section button and press the other (**ACCENT** + **SLIDE**): the slot's A and B sections link into one pattern that plays A then B, up to 64 steps. The link is stored with the pattern, so it comes back linked whenever the pattern is selected, and a linked pattern always starts from its A section. Both section LEDs light, with the playing or edited section blinking. **CLEAR** on a linked pattern erases both halves. Writing pitches into a linked pattern fills section A first, then hands the cursor to section B. To split the pair again, press **ACCENT** or **SLIDE** alone.

C. GROUPS

Turn the Group Dial to browse another group while running, then press a pattern key: the group and pattern switch together at the next wrap.

D. TEMPO LED

While the sequencer is stopped, the selected pattern LED keeps flashing at the tempo (120 BPM idle, or the incoming clock tempo), like a real 303.

1 STEP EDITING

MODE DIAL **PATTERN WRITE** · SEQUENCER STOPPED

Step editing walks the pattern one step at a time so you can fix a single note without rewriting the rest. Enter **PITCH MODE** or **TIME MODE**, then hold **WRITE/NEXT**: the current step is auditioned and can be edited for as long as the key is held.

A. EDITING PITCH

- Press a keyboard key to replace the step's pitch; the step re-auditions.
- **UP** / **DOWN** nudge the octave.
- **ACCENT** / **SLIDE** toggle the step's flags.
- Release **WRITE/NEXT** to advance to the next note step (rests and ties are skipped). **BACK** steps backward.

B. EDITING TIME

In **TIME MODE** with **WRITE/NEXT** held, press **STEP**, **TIE**, or **REST** to change the current step's type, or **LOCK** to toggle its lock.

> **NOTE**
>
> To jump straight to a specific step instead of walking to it, use Step-Select Editing (Performance, section 2).

2 PATTERN OPERATIONS

MODE DIAL **PATTERN WRITE** · HOLD **CLEAR**

Holding **CLEAR** turns the panel into a pattern-transform toolkit. None of these operations move the playhead or interrupt the clock, so they are safe to use mid-performance. They apply to whichever variation is the current edit target (Variations & Scales, section 1).

A. TRANSFORM

WHILE HOLDING CLEAR	RESULT
ACCENT	Randomize the pattern
SLIDE	Mutate: a small random perturbation
DOWN / UP	Rotate time data one step left / right
PITCH MODE + DOWN / UP	Rotate pitch data one step left / right
BACK / WRITE/NEXT	Shift pitch and time one step left / right
F#	Reverse the pattern within its length
G#	Clear pitches only (keep time data)
A#	Clear time data only (keep pitches)

Random and mutate operations respect the pattern's scale, if one is enabled (Variations & Scales, section 3).

B. RANDOMIZE ONE ATTRIBUTE

Hold CLEAR + PITCH MODE together and tap a white key:

C	D	E	F	G	A
Semitones	Octaves	Accents	Slides	All pitch data	Time data

C. COPY AND PASTE

Hold CLEAR + C# and press a pattern key to copy that slot to the clipboard. Hold CLEAR + D# and press a pattern key to paste into that slot. Both operate within the current bank.

3 CHAINING PATTERNS

MODE DIAL PATTERN PLAY

A chain plays up to four patterns in sequence, wrapping back to the first. A chain entry that is a linked A/B pair plays its A and B sections before the chain advances, and chains always enter a linked pair at its A section. Stopping and restarting the sequencer replays the chain from its first entry.

A. BUILDING A CHAIN (STOPPED)

Hold the first pattern key as an anchor and press up to three more keys in the same bank. Release the anchor to activate the chain. Releasing the anchor with only one key held deactivates any existing chain.

B. BUILDING A CHAIN IN ANY ORDER (A + B HELD)

Hold ACCENT + SLIDE together and tap up to four pattern keys in any order; repeats are allowed (1 1 2 4 is a valid chain). Releasing the buttons commits it: stopped, it starts immediately; running, it takes over at the next pattern wrap.

C. WHILE RUNNING

- A short tap (under 300 ms) on a pattern key queues that pattern next. With a chain playing, the chain finishes its current pass first, then hands over.
- Holding a pattern key (300 ms or more) loops that pattern for as long as it is held.
- Pressing two pattern keys together, or holding one and tapping another, builds a new chain from the range between them. With a chain already playing, the new chain is queued and takes over at the chain boundary.

D. SAVING A CHAIN WITH A PATTERN

A chain can be stored on its first pattern so it comes back later:

- While the chain-build keys are still held, press FUNCTION (Pattern Write), or
- while holding ACCENT + SLIDE, press WRITE/NEXT to save the build in progress or the currently playing chain.

The mode LEDs flash to confirm. Selecting that pattern later while stopped (or powering on with it selected) re-arms the whole chain automatically; a pattern without a stored chain plays alone. ACCENT + SLIDE + WRITE/NEXT with no chain active clears the stored chain from the current pattern, and clearing the pattern clears its stored chain with it.

4 PLAYBACK DIRECTION

Press FUNCTION + TIME MODE to open the direction selector; six LEDs show the available directions. Press a key to choose, then FUNCTION to exit.

KEY	DIRECTION
C	Forward
D	Reverse
E	Ping-pong: forward, then backward
F	Random: every step chosen at random
G	Half random: a chance that the current step is replaced by a random one
A	Brownian: a chance of stepping backward or jumping ahead, drifting through the pattern

While running, a direction change is queued and takes effect at the next pattern wrap, so the pattern always hands off in phase with the clock. Each variation stores its own direction (Variations & Scales, section 1). Slides are direction-aware; random modes never slide.

5 METRONOME TAP-WRITE

MODE DIAL PATTERN WRITE · SEQUENCER RUNNING

Metronome tap-write records rhythm the way you would tap it on a table, exactly like the original 303's TAP timing method, with the original's sound: a short click on every 8th note and a lower click marking the start of each bar. The recorder's timing windows are measured cycle-exact against the original mask ROM, so taps land on the same steps they would on a stock 303. Press CLEAR + TIME MODE to switch it on: the pattern's time data is cleared, its pitches are kept and consumed in order as you tap, and the bar restarts from step 1.

- Tap WRITE/NEXT on a beat to write a note there; a tap in the later part of a step aims at the next step's downbeat, so slightly early taps land where you meant them.
- Holding the key across further beats writes ties; releasing it ends the note.
- Rests are never written over earlier passes, so each pass adds on top of what is already there.
- Taps aimed past the end of the bar are dropped, like the original's "do not tap between two measures".
- You hear each tapped note from the instant you press, at the pitch it will consume, until you let go.
- **Sustain**: hold any pattern key 1–8 and the gaps between notes fill with ties instead of rests.

A. AN ENDLESS SESSION

Unlike the original, which stops after one measure, the session loops until you press CLEAR + TIME MODE again, stop the sequencer, or leave Pattern Write. While the pattern is empty the metronome keeps clicking and the voice stays silent (**record**). Once a bar wraps with notes in it, the clicks stop and the pattern plays back cleanly while your taps keep recording on top (**overdub**). Emptying the pattern brings the metronome back. Chained patterns join the session as they come around, each keeping its own record/overdub state.

> **NOTE**
>
> During a session the pattern keys belong to the recorder (sustain), and the CLEAR transform combos are disabled until the session ends.

6 STEP PROBABILITY

MODE DIAL PATTERN WRITE · VARIATION 1 ONLY

Step probability turns fixed steps into ones that only sometimes happen, so a line breathes and never repeats the same way twice. Any step can arm four independent characteristics: an **accent**, a **slide**, an octave **down** (–12), and an octave **up** (+12, or +24 for double-up). Each carries its own probability, and each is rolled fresh every time that step plays. A tie holds the previous step's roll; a rest clears it. Probability rides **variation 1** (the analog voice) only.

Press FUNCTION + SLIDE to open the editor, and FUNCTION to exit. Steps that carry probability light bright; empty note steps show dim for orientation, and the playhead chases across the visible bank while running.

A. SELECTING A STEP

Black keys pick which bank of eight steps you are looking at (C# 1–8 through G# 25–32); white keys 1–8 select a step within that bank. BACK deselects. The selected step's LED flashes. On a linked A/B pair, each section carries its own probability: select the section first, then edit its steps.

B. ARMING CHARACTERISTICS

With a step selected (and WRITE/NEXT not held):

KEY	RESULT
ACCENT	Toggle a probabilistic accent
SLIDE	Toggle a probabilistic slide
DOWN	Toggle an octave-down shift (–12)
UP	Cycle none → up (+12) → double-up (+24)

Arming a characteristic sets it to 100% (always happens) as a starting point. A characteristic can be armed on a rest or tie step too; it stays latent and applies if that step later becomes a note.

C. SETTING THE ODDS

Hold WRITE/NEXT to enter the level layer. Tap ACCENT, SLIDE, DOWN, or UP to pick which characteristic you are editing (its LED blinks), then press a keyboard key C through high C to set the level 1–13. The pitch LEDs show the level as a bar. Level 13 is certain (100%); level 1 is the rarest, roughly one step in thirteen.

> **NOTE**
>
> Down and up roll independently. If both pass on the same step, the sequencer flips a coin to decide which shift applies, so a step can never move two directions at once.

D. RANDOMIZING

Hold CLEAR and tap ACCENT, SLIDE, DOWN, or UP to scatter random odds for that one characteristic across the whole pattern. CLEAR + WRITE/NEXT randomizes all four at once. About a third of the steps land unarmed and the rest get a random level.

Variations 2 and 3 carry no probability. Probability edits are mirrored to the web pattern editor's PROB lane, and vice versa.

1 THREE VARIATIONS PER PATTERN

Every pattern slot holds **three independent sequences** that play simultaneously, so one slot can drive the 303 line plus two external synth lines in lockstep:

VARIATION	OUTPUT	MIDI CHANNEL
1	The 303 analog voice + MIDI	Main channel
2	MIDI only	Own channel (default 2)
3	MIDI only, mono or polyphonic	Own channel (default 3)

Each variation has its own notes, length, direction, transpose, and scale.

A. PICKING WHICH VARIATION YOU EDIT

1. Mode Dial to Pattern Write, normal mode (no FUNCTION / PITCH MODE / TIME MODE held).
2. Hold WRITE/NEXT. The picker appears on C / D / E: the selected variation flashes, the other two show dim.
3. Still holding WRITE/NEXT, press C, D, or E to select variation 1, 2, or 3.

All editing (pitch and time modes, step-select, CLEAR combos, direction, scale) then applies to the selected variation, and the chase LEDs follow its playhead. The selection returns to variation 1 when you change patterns.

B. SETTING THE VARIATION MIDI CHANNELS

In Menu Mode (FUNCTION + CLEAR): hold PITCH MODE while pressing a channel key to set the variation-2 channel, or hold SLIDE for variation 3. Without a modifier the keys set the main channel as before (Performance, section 6).

2 VARIATION 3 POLYPHONY

Variation 3 can play **chords of up to four notes per step** over MIDI. The poly flag is stored per pattern slot.

A. TURNING POLY ON / OFF

1. Stop the sequencer.
2. Hold WRITE/NEXT (variation picker) and select variation 3 with E.

3. Still holding WRITE/NEXT, press SLIDE. The SLIDE LED shows the state: lit = polyphonic, off = monophonic.

B. WRITING CHORDS

1. With variation 3 (poly) selected as the edit target, stop the sequencer and press PITCH MODE (Pattern Write) to open the chord editor.
2. Note keys toggle chord notes on the current step; each press auditions the chord over MIDI.
3. Tap UP / DOWN to set the octave new notes land in (it latches).
4. TIME MODE cycles the step between note / tie / rest. ACCENT and SLIDE set per-step flags.
5. WRITE/NEXT / BACK move to the next / previous step, with audition.

3 PER-PATTERN SCALES

MODE DIAL PATTERN WRITE

Each pattern can carry a **scale filter**: notes are pulled to the nearest allowed pitch at playback. The filter is non-destructive; your stored notes are untouched, and switching the scale off restores them exactly. Random and mutate operations also respect it.

1. Hold FUNCTION and press ACCENT to enter scale mode.
2. Note keys toggle the twelve pitch classes (lit = allowed). The high C′ key toggles the scale on / off for this pattern.
3. Presets: hold FUNCTION and press a note key to load a preset rooted on that note; pressing the same note again cycles through the 35 preset scales (major, minors, modes, pentatonics, blues, and more).
4. Tap FUNCTION (press and release with no note) to exit.

1 LIVE PATTERN EDIT

MODE DIAL PATTERN WRITE · SEQUENCER RUNNING

With the sequencer running, press PITCH MODE: anything you play is recorded into the pattern as it passes. Tap keyboard keys to write pitches, and UP / DOWN, ACCENT, or SLIDE to write octaves and flags.

The same works for rhythm: press TIME MODE while running and tap STEP, TIE, REST, or LOCK to record them live. Locked steps ignore live input, so use LOCK to protect the parts you want to keep. Press FUNCTION to exit.

2 STEP-SELECT EDITING

MODE DIAL PATTERN WRITE · RUNNING OR STOPPED

Step-select jumps straight to any step of a pattern and edits it in place while the pattern keeps playing, uninterrupted and unmuted. Press FUNCTION + PITCH MODE to enter. The pattern keys become a step overview: lit = note, dim = tie, unlit = rest. The black keys jump between 8-step banks (C# 1–8, D# 9–16, F# 17–24, G# 25–32). On a linked A/B pair, steps 33–64 are the B section: select section B first, then edit its steps the same way.

A. EDITING A STEP'S PITCH

Press a lit step to select it (it blinks), then WRITE/NEXT to open it. Keyboard keys set the pitch, UP / DOWN nudge the octave, ACCENT / SLIDE toggle flags, and WRITE/NEXT auditions it. The playhead is not disturbed. Press BACK to return to the overview.

B. EDITING A STEP'S TIME

Press TIME MODE to switch to the time view. Select any step, then press STEP, TIE, or REST directly; no need to open the step. Press FUNCTION to exit step-select.

C. WITH A CHAIN ACTIVE

When a chain of two or more patterns is active, DOWN / UP / ACCENT / SLIDE switch which chained pattern you are viewing (in the time view, only while no step is selected). The LEDs show the viewed slot solid, the playing slot as a bright chase blink, and the others dim.

NOTE

If variation 3 (poly) is the edit target, step-select edits the chord stream instead: the detail editor works on the selected step's chord.

3 KEYBOARD PLAY MODE

MODE DIAL PATTERN PLAY · RUNNING OR STOPPED

Play the 303's analog voice live from the panel keys. Hold FUNCTION and press PITCH MODE to toggle the mode.

- Note keys play the voice; overlapping presses slide (legato), and releasing slides back to a still-held key, like a mono synth.
- Octaves: the TIME MODE key toggles octave latch (TIME MODE LED lit = latched, the default). Latched: tap DOWN / UP to step through octaves 0–3 and it stays. Unlatched: hold DOWN / UP like the stock transpose.
- Everything you play also goes out over MIDI per key, so a mono synth slides and a DAW records true note lengths.
- Press FUNCTION to leave the mode.

4 TRANSPOSE & FORCE ACCENT / SLIDE

A. LIVE TRANSPOSE

Hold PITCH MODE (running or stopped, outside Pattern Write) and tap any keyboard key to transpose the pattern to that root; each step's octave is preserved. With PITCH MODE still held, UP / DOWN shift the transpose octave. Release to exit. Transpose is non-destructive and resets when you switch patterns; the stored pattern is unchanged.

B. FORCE ACCENT / SLIDE

MODE DIAL PATTERN PLAY / TRACK PLAY · SEQUENCER RUNNING

While holding PITCH MODE, also hold ACCENT to force an accent on every step, or SLIDE to force slides. Force-slide also holds the gate across every non-rest step for continuous portamento. The effect lasts while held and reverts on release. It is applied at the analog output stage only; the pattern is not modified.

5 THE TRACK

A track is a song: a sequence of up to 64 bars, each bar naming one pattern and an optional transpose. There are eight track slots, selected with the Group Dial in either track mode. Tracks 1–2 use patterns from group I, 3–4 from group II, 5–6 from group III, and 7–8 from group IV.

A. WRITING A TRACK

MODE DIAL **TRACK WRITE** · SEQUENCER RUNNING

1. Choose a track with the Group Dial and press **RUN/STOP**.
2. Select the pattern to record: pattern keys **1**–**8**, with **ACCENT** / **SLIDE** choosing section A / B.
3. Press **WRITE/NEXT** to write the playing pattern into the current bar and move to the next.
4. For the final bar, press **CLEAR** and then **WRITE/NEXT**: the bar is written, marked as the end, and the cursor returns to bar 1.
5. Stop the sequencer to save the track.

B. REVIEWING (STOPPED)

While stopped, **WRITE/NEXT** and **BACK** step through the written bars, auditioning each bar's pattern. **CLEAR** returns to bar 1. Writing only happens while the clock is running, so browsing is non-destructive.

C. PER-BAR TRANSPOSE

In Track Write, hold **PITCH MODE**: the keyboard sets the current bar's transpose, and **UP** / **DOWN** choose its octave. The pitch and octave LEDs display the stored value while held.

D. PLAYING A TRACK

MODE DIAL **TRACK PLAY**

Choose the track with the Group Dial and press **RUN/STOP**. The track advances one bar at every pattern wrap and loops at the end. While stopped, **CLEAR** resets the playhead to bar 1.

6 MENU MODE

Press **FUNCTION** + **CLEAR** to open the settings menu. Everything is visible and editable at once on a single combined screen. Press **CLEAR** or **FUNCTION** to exit.

CONTROL	SETTING
Pattern keys 1 – 8	MIDI channel. Low bank = channels 1–8, high bank = 9–16.
D#	High-bank latch for the channel keys. LED on while latched.
Hold PITCH MODE + channel key	Variation-2 MIDI channel.
Hold SLIDE + channel key	Variation-3 MIDI channel.
TIME MODE	Clock source. LED on = DIN sync, off = MIDI clock.
ACCENT	MIDI output mode. LED on = MIDI OUT, off = MIDI THRU.
A# + UP / DOWN	LED brightness (1–8), saved to memory.
G#	Switch firmware, superOS ↔ D650C (Combined Edition, section 1).

1 TWO FIRMWARES IN ONE BOX

The Combined Edition holds two complete operating systems and boots one of them at a time:

- **superOS-303**: the modern OS described in this manual. 64-step patterns, variations, polyphony, scales, MIDI, web editor.
- **D650C**: a bit-perfect emulation of the original TB-303 CPU running the genuine Roland mask ROM. It behaves exactly like a stock 1982 TB-303, including the original write workflow and quirks.

A. SWITCHING FIRMWARE

1. Open Menu Mode: hold FUNCTION and press CLEAR (works in both firmwares).
2. Look at the G# LED: solid = superOS is running, blinking = D650C is running.
3. Press G#. The unit saves everything, waits until all buttons are released, and reboots into the other firmware in about a second.

Both memories are kept: superOS patterns live in flash, D650C patterns in a separate memory. Switching back and forth never erases either side. A watchdog guards the boot and the swap, so a failed start recovers on its own; if the unit ever appears stuck, power-cycle it, the selection is already saved and it will boot into the chosen firmware.

> **NOTE**
>
> Keep your hands off the buttons while it swaps. The firmware waits for a quiet panel before rebooting, so a held key simply delays the swap by a moment.

2 D650C MODE

In D650C mode the unit *is* a stock TB-303: use the original owner's-manual workflow (pattern clear, pitch mode, time mode with the tap metronome, tracks, chaining, transpose). Nothing from superOS applies here except the additions below.

A. LOADING THE D650C MASK ROM

The emulator runs the genuine Roland TB-303 CPU mask ROM. That ROM is copyrighted, so the public firmware ships **without** it: you supply your own 2 KB dump (read from your own chip, or a dump you are legally entitled to use) and load it into the unit once. It is kept in permanent on-chip memory and survives firmware updates done over MIDI, so this is a one-time step. Some units come with the ROM already built in, in which case D650C plays immediately and you can skip this.

No ROM yet? When you switch into D650C mode with no ROM installed, the unit waits for one: the low C and high C key LEDs blink back and forth, slowly while it waits and quickly while a ROM is transferring. The web editor's header also shows *no mask ROM* in red.

1. Open the **web editor** and connect it to the unit over MIDI using a USB-to-DIN MIDI interface.
2. In the **D650C mask ROM** strip, click **Load ROM dump** and pick your file: a raw 2048-byte `.bin`, or a converted `.syx` file. The status line confirms the size when it loads.
3. Make sure the unit is in D650C mode (with no ROM it is already waiting there), then click **Send to device.**
4. The key LEDs switch to a fast blink and stay there for the whole operation. The transfer itself is quick (a second or so); most of the wait is the unit writing the ROM into permanent memory, which takes several seconds. When it finishes it reboots straight into the emulator. Done.

> **NOTE**
>
> You can also click
>
> SAVE .SYX
>
> to convert a `.bin` dump into a `d650c-maskrom.syx` file and send it with any SysEx tool (SysEx Librarian, MIDI-OX) while the unit is in D650C mode. The ROM is checksummed on arrival; a bad or interrupted transfer is rejected and the previously installed ROM, if any, is kept.

B. PATTERN MEMORY

The original battery-backed pattern RAM is replaced by permanent on-chip storage. Patterns save automatically in the background (a second or two after you stop editing or playing) with zero playback stutter. No battery, nothing to do.

C. CONFIG MENU (FUNCTION + CLEAR)

CONTROL	SETTING
1 – 8 (+ D#)	MIDI channel 1–8; tap D# first (LED lit) for channels 9–16.
TIME MODE	Cycle clock source: internal → MIDI clock → DIN sync. TIME LED lit = external source; PITCH LED also lit = DIN sync.
ACCENT	Toggle MIDI soft-thru.
A# + UP / DOWN	LED brightness 1–8.
G#	Switch to superOS (blinking = D650C is the one running now).
CLEAR or FUNCTION	Exit the menu.

D. MIDI IN D650C MODE

- Note out: the sequencer is mirrored to MIDI out with slide legato and accent as velocity 127.

- Note in: while stopped, incoming notes play the 303 voice live.
- Clock: slaves to MIDI clock or DIN sync (select in the menu); with the internal clock it sends MIDI clock out.
- External transport (Start / Stop or the DIN run line) starts and stops the sequencer exactly like the original's sync jack.

3 PITCH CALIBRATION

Both firmwares use the factory pitch standard from the Roland service notes: the untransposed low C key puts exactly 1.000 V on the CV jack. superOS and D650C play identical pitches for the same key, and MIDI is aligned the same way in both (low C key = MIDI note 48 / C3).

COMING FROM AN OLDER SUPEROS BUILD?

Older superOS output sat one semitone above the factory spec. If your instrument was tuned against it, everything will now sound one semitone flat until you turn the front-panel TUNING knob up one semitone, once. After that, both firmwares match each other and a factory-calibrated TB-303.

1 MIDI & DAW CONTROL

MIDI runs over the DIN jacks, in both directions:

- Sequencer notes out (all three variations on their channels), clock and transport in / out.
- Live note input while stopped plays the 303 voice; velocity 100 or higher adds accent, and legato playing slides.
- Pattern remote control: CC 0 selects the group (0–3), CC 32 the section (0 = patterns 1–8, 1 = 9–16), and a Program Change selects the pattern, matching a DAW's bank / program menus.

2 WEB PATTERN EDITOR

A browser-based pattern editor is available at **superos303.com/editor** and as a standalone file in `tools/web-pattern-edit/index.html` in the repository. Connect over WebMIDI (requires HTTPS or localhost). The editor receives all pattern data from the hardware over SysEx; changes sync both ways instantly, including edits made on the panel.

- Edit all 16 patterns of a group: pitch, accent, slide, time (note / tie / rest), step lock, length up to 64 steps (24 in triplet mode), direction, and per-pattern scale.
- All three variations per slot, including the variation-3 chord editor when poly is on, plus the variation MIDI channel and device configuration.
- Pattern chains and track editing with live track telemetry.
- Live playhead chase while the sequencer runs, derived from the incoming MIDI clock and re-anchored at every pattern wrap, so position tracking costs no extra SysEx traffic.
- Shows which firmware is running and its version, and can switch the unit between superOS and D650C.
- Back up all patterns to a file and restore them, recommended before any firmware update.

3 PATTERN STORAGE

superOS patterns, tracks, and settings live in a large wear-levelled area of chip flash rather than the tiny hardware EEPROM. There is no manual save step:

- Saves happen on transport stop and about two seconds after you stop editing, on the panel or in the web editor.
- Every pattern slot stores all three variations and, where enabled, the variation-3 chord data.
- Patterns, tracks, and settings survive firmware updates done over MIDI, except for the occasional update that moves the storage layout. v1.0-beta is one of those, so back up before installing it (see

Firmware Update).

- D650C mode keeps its own separate pattern memory; the two never overwrite each other.

4 FIRMWARE UPDATE

Updates are one SysEx file over MIDI; no programmer needed. Release files are self-contained, so any SuperOS-303 CPU with the bootloader can take them.

1. Enter the bootloader: hold TAP (WRITE/NEXT) while powering on, or send SysEx `F0 7D 4A F7` to the running unit (works in both firmwares). Four LEDs light solid: the device is in the bootloader.
2. Send the release `.syx` file over MIDI with a SysEx tool that supports a delay between messages (e.g. SysEx Librarian, ~50 ms).
3. The unit reboots into normal operation by itself when the file ends.

If the upload fails and the LEDs stay stuck, it is safe to power-cycle and try again. Stored patterns are never touched by the bootloader.

CAUTION

Back up your patterns over SysEx before updating, and do not disconnect power mid-update unless the update has already failed.

UPDATING TO V1.0-BETA MOVES THE PATTERN STORAGE LAYOUT (FROM V0.9.9C OR ANY EARLIER VERSION), SO ITS FIRST BOOT STARTS WITH EMPTY SUPEROS PATTERNS.

Back up with the web editor first and restore afterwards. Updates that do not move the layout keep everything. D650C patterns and any installed mask ROM are stored separately and are not affected. See Precautions at the front of this manual.

5 QUICK REFERENCE

WRITING · PATTERN WRITE

Hold pattern key + CLEAR	Erase that pattern slot
PITCH MODE / TIME MODE	Write pitches / rhythm
FUNCTION + STEP (taps)	Count out the pattern length
FUNCTION + black key, white key	Set length directly (section, to 32)
FUNCTION + SLIDE, then step	Length into the B section (33–64, links the pair)
FUNCTION + WRITE/NEXT / BACK	Length +1 / –1
FUNCTION + UP / DOWN	Double / halve the length
TIME MODE: FUNCTION + UP	Triplet mode on / off
Hold WRITE/NEXT (in pitch / time mode)	Step edit (release = next step)
Hold WRITE/NEXT (normal mode) + C / D / E	Pick edit variation 1 / 2 / 3
… + E, then SLIDE (stopped)	Variation 3 poly on / off (LED lit = poly)
FUNCTION + ACCENT	Scale mode (FUNCTION + note = presets)

TRANSFORM · HOLD CLEAR

+ ACCENT / SLIDE	Randomize / mutate pattern
+ DOWN / UP	Rotate time data left / right
+ PITCH MODE + DOWN / UP	Rotate pitch data left / right
+ BACK / WRITE/NEXT	Shift everything left / right
+ F# / G# / A#	Reverse / clear pitches / clear time
+ PITCH MODE + C … A	Randomize one attribute (see Pattern Editing 2)
+ C# / D# + pattern key	Copy / paste pattern
+ TIME MODE (running, Pattern Write)	Metronome tap-write on / off

■ PERFORMANCE

Pattern key (tap / hold, running)	Queue pattern / loop that pattern
Hold pattern key + more keys	Build a chain (up to 4)
Hold ACCENT + SLIDE + pattern keys	Build a chain in any order (repeats allowed)
… + WRITE/NEXT, or build + FUNCTION	Save the chain on its first pattern (recalled on select)
ACCENT / SLIDE	Section A / Section B (unlinks a pair)
Hold one section button + press the other	Link A + B into one 64-step pattern
Hold PITCH MODE + key	Transpose (+ UP / DOWN = octave)
Hold PITCH MODE + ACCENT / SLIDE	Force accent / slide (running)
FUNCTION + PITCH MODE (Pattern Write)	Step-select editing
FUNCTION + PITCH MODE (Pattern Play)	Keyboard play mode (TIME MODE = octave latch)
FUNCTION + TIME MODE	Playback direction selector
FUNCTION + SLIDE (Pattern Write)	Step probability editor (variation 1)

■ TRACK · TRACK WRITE / TRACK PLAY

WRITE/NEXT (running)	Write current pattern to the bar, advance
CLEAR, then WRITE/NEXT	Write the final bar, return to bar 1
Hold PITCH MODE + key	Set the bar's transpose
WRITE/NEXT / BACK (stopped)	Review bars forward / backward
CLEAR (stopped)	Back to bar 1

■ SYSTEM

FUNCTION + CLEAR	Menu mode (MIDI channels, sync, brightness)
… then G#	Switch firmware, superOS ↔ D650C (solid = superOS, blinking = D650C)
… hold PITCH MODE / SLIDE + channel key	Variation 2 / 3 MIDI channel
Hold TAP at power-on, or SysEx `F0 7D 4A F7`	Bootloader for firmware update

6 SPECIFICATIONS

■ **PATTERNS**

64 slots: 4 groups × 8 patterns, each with A and B sections of 1–32 steps (1–24 in triplet mode). A linked A/B pair plays as one pattern of up to 64 steps, and the link is stored with the pattern. Per step: pitch over a four-octave range, accent, slide, live step lock.

■ **VARIATIONS**

3 independent sequences per pattern slot, playing simultaneously: variation 1 on the analog voice + MIDI, variations 2 and 3 on their own MIDI channels. Variation 3 optionally polyphonic with chords of up to 4 notes per step. Per-variation length, direction, transpose, and scale.

■ **SCALES**

Per-pattern non-destructive scale filter over the 12 pitch classes, with 35 presets (major, minors, modes, pentatonics, blues, and more).

■ **PLAYBACK**

Six directions: forward, reverse, ping-pong, random, half random, Brownian. Chains of up to 4 patterns, saveable per pattern and recalled on select or power-on. Non-destructive transpose and force accent / slide. Keyboard play mode with legato slide. Metronome tap-write with looping record/overdub session, timing-matched cycle-exact to the original mask ROM.

■ **PROBABILITY**

Per-step probability on variation 1: independent accent, slide, octave-down, and octave-up (or double-up) characteristics, each with its own 1–13 level, rolled fresh every step.

■ **TRACKS**

8 track slots (2 per group), up to 64 bars each, with per-bar pattern and transpose.

■ **MIDI**

DIN MIDI, in and out. Three note channels (main + variation 2 + variation 3). Clock sync from MIDI or DIN sync. Switchable MIDI OUT / THRU. Pattern remote control via CC 0 / CC 32 / Program Change. SysEx pattern transfer for the web editor and backups; firmware updates and firmware version reporting over SysEx.

■ **MEMORY**

Patterns, tracks, and settings in wear-levelled on-chip flash, retained without power and across firmware updates. Automatic background saves on stop and after edits; no manual save step.

■ **COMBINED EDITION**

Dual-boot image: superOS-303 plus a cycle-accurate D650C emulator running the genuine TB-303 mask ROM, each with its own pattern memory. Panel or SysEx firmware switching with watchdog-guarded boot.

■ **HARDWARE**

VOX DEI CPU by Michigan Synth Works (AT90USB1286, Teensy++ 2.0 compatible), a drop-in replacement for the TB-303's original CPU. Factory pitch standard: low C = 1.000 V CV, MIDI note 48.

■ LICENSE

Open source under the MIT License. Fork of DJ Phazer's OS-303.

SUPEROS-303 FIRMWARE OWNER'S MANUAL, COMBINED EDITION V1.0-BETA. FORK OF DJ PHAZER'S OPEN-SOURCE OS-303 (MIT LICENSE), BUILT FOR THE VOX DEI CPU BY NICHOLAS MICHALEK / MICHIGAN SYNTH WORKS. D650C MODE RUNS THE GENUINE TB-303 MASK ROM VIA A CYCLE-ACCURATE UCOM-43 CORE.